From the Fig.7.30, we see that

Volume of a sphere $=$ $2 \times$ (Volume of a cone)

where the diameters of sphere and cone are equal to the height of the cone.

$$= 2\left(\frac{1}{3}\pi r^2 h\right)$$

$$= \frac{2}{3}\pi r^2 \left(2r\right), \quad (\because h = 2r)$$

Volume of a sphere $= \frac{4}{3}\pi r^3$ cu. units

7.3.5 Volume of a hollow sphere / spherical shell (volume of the material used)

Let r and R be the inner and outer radius of the hollow sphere.

Volume enclosed between the outer and inner spheres

$$= \frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3$$

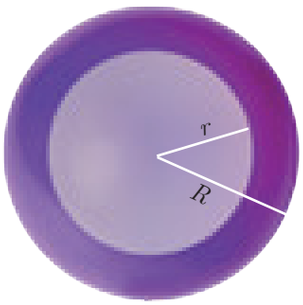


Fig. 7.31

Volume of a hollow sphere $= \frac{4}{3}\pi(R^3 - r^3)$ cu. units

7.3.6 Volume of solid hemisphere

Let r be the radius of the solid hemisphere.

Volume of the solid hemisphere $= \frac{1}{2}$ (volume of sphere)

$$= \frac{1}{2}\left[\frac{4}{3}\pi r^3\right]$$

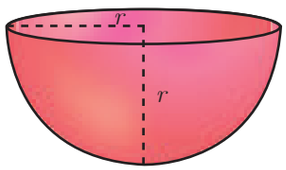


Fig. 7.32

Volume of a solid hemisphere $= \frac{2}{3}\pi r^3$ cu. units

7.3.7 Volume of hollow hemisphere (volume of the material used)

Let r and R be the inner and outer radius of the hollow hemisphere.

$$\left.\begin{matrix}\text{Volume of hollow} \\ \text{hemisphere}\end{matrix}\right\} = \left[\begin{matrix}\text{Volume of outer} \\ \text{hemisphere}\end{matrix}\right] - \left[\begin{matrix}\text{Volume of inner} \\ \text{hemisphere}\end{matrix}\right]$$

$$= \frac{2}{3}\pi R^3 - \frac{2}{3}\pi r^3$$

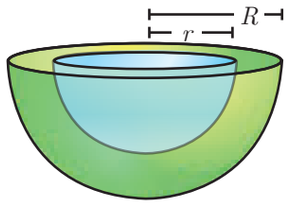


Fig. 7.33

Volume of a hollow hemisphere $= \frac{2}{3}\pi(R^3 - r^3)$ cu. units

Thinking Corner

A cone, a hemisphere and a cylinder have equal bases. The heights of the cone and cylinder are equal and are same as the common radius. Are they equal in volume?

7 Mensuration.indd 287 16-12-2020 03:18:05 PM